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foundationally, adversarially and from first-principles attack this whitepa

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foundationally, adversarially and from first-principles attack this whitepaper I'm busy writing. the intent should be clear.

Attack the mathematics AND the physics. I'm both a

Show more

I'll treat this as a hostile technical referee report: isolate the core claim chain, separate proved/imported/open material, and attack definitions, normalizations, hidden assumptions, and overclaims.

The main vulnerability is sharper now: the determinant/K_0 package may be clean but tautological, while the physics language risks upgrading a recognition ledger into realization.

Thought for 11m 9s

I'm going to be brutal. The draft is strongest as a **normalization-and-separation paper**: it carefully distinguishes the Borcherds product, the Gritsenko–Nikulin denominator algebra, the OP/DT scalar square, and an open compact Hall/factorization realization. It is weakest whenever the prose makes the reader feel that these layers are closer than they are. The main danger is not that the automorphic mathematics is fake; the main danger is that the physical and Hall-theoretic interpretation is not yet mathematically supported by an additive charge category.

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Executive verdict

The paper's core theorem is basically:

$$\phi_{0,1} \overset{\text{Borcherds/Gritsenko–Nikulin}}{\qquad\qquad} \Delta_5$$

with a careful convention that the theta-normalized form has leading coefficient 64, so that the monic product is $D_5 = 64^{-1}\Delta_5$. That part is fine and externally anchored: Gritsenko–Nikulin explicitly write $64^{-1}\Delta_5$ as the product with exponents $f(nm, l)$, the Fourier coefficients of $\phi_{0,1}$. arXiv +1

But the paper risks sounding like it has built a **BPS square root** for $K3 \times E$. It has not. It has built a **virtual K_0-determinant** and aligned it with an existing BKM denominator theorem. Your own draft repeatedly admits this: the half-index object is only a $K_0(\mathrm{sVect})$ representative with no canonical Hilbert-space half, no parity lift, and no operator product; the compact Hall/chiral realization is explicitly open.

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The single most serious foundational problem is this:

$$(n, l, m)$$

is being used in three incompatible ways: Fourier exponent, Lorentzian BKM root degree, and would-be physical/Hall charge degree. The first two are additive. The third is generally **not** additive if (n, l, m) are Gram invariants of physical charges.

That has to be fixed before any compact Hall/factorization story can be more than a recognition wish-list.

1. Fatal issue: Gram invariants are not physical Hall charges

You define

$$\Gamma_{\text{gram}} = \Gamma_{\text{ind}} = \mathbb{Z}^3$$

and explicitly say it is "notation for Gram invariants and Fourier indices," not a microscopic charge lattice. Then later the compact Hall discussion assumes an extension-closed Hall category with numerical Gram-index lattice Γ_{ind}, semistable stacks indexed by $\gamma \in \Gamma_{\text{ind}}$, and Hall correspondences landing in $\gamma + \eta$.

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That is not a small technical gap. It is structurally wrong unless you supply a new construction.

If physical charges are electric-magnetic pairs (Q, P), then

$$n = \frac{Q^2}{2}, \qquad l = Q \cdot P, \qquad m = \frac{P^2}{2}.$$

But for two charge pairs,

$$\frac{(Q_1 + Q_2)^2}{2} = \frac{Q_1^2}{2} + \frac{Q_2^2}{2} + Q_1 \cdot Q_2,$$

$$(Q_1 + Q_2) \cdot (P_1 + P_2) = Q_1 \cdot P_1 + Q_2 \cdot P_2 + Q_1 \cdot P_2 + Q_2 \cdot P_1,$$

$$\frac{(P_1 + P_2)^2}{2} = \frac{P_1^2}{2} + \frac{P_2^2}{2} + P_1 \cdot P_2.$$

So the map

$$(Q, P) \mapsto (n, l, m)$$

is quadratic. It does **not** send physical charge addition to addition in $\mathbb{Z}^3$, except in special cross-orthogonal situations. A Hall algebra must be graded by an additive K-theory, Mukai, or electromagnetic charge lattice. It cannot be fundamentally graded by the Gram invariants unless you prove a quotient/descent theorem saying that the Hall product descends through this nonlinear invariant map.

The same issue appears on the D6–D2–D0 side. You define rank-one charges

$$\gamma(h, d, n) = (1; (\beta_h, d); n), \qquad \beta_h = s_1 + hF,$$

and the scalar DT series sums over h, d, n. But the rank-one D6 sector is not extension-closed under Hall multiplication: extensions/direct sums of rank-one objects naturally leave the rank-one sector. Also $h - 1 = \beta^2/2$ is quadratic in β, not additive. The family $\beta_h = s_1 + hF$ is not closed under addition in the way a Hall grading would require.

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What a hostile referee will say

"You have a beautiful additive root lattice for the automorphic product, but you have not shown that it is the additive charge lattice of the compact BPS category. You have shown an additive algebra of Fourier exponents, not an additive algebra of physical charges."

Required repair

Introduce two lattices, not one:

$$\Gamma_{\text{phys}} \quad \text{and} \quad \Gamma_{\text{gram}}.$$

The Hall category must be graded by Γ_{phys}. Then define a non-additive invariant map

$$\pi : \Gamma_{\text{phys}} \to \Gamma_{\text{gram}}.$$

The automorphic product is the π-pushed-forward index. If you want a primitive Lie algebra graded by Γ_{gram}, you must prove a descent theorem:

$$\mathrm{Prim}_{\Gamma_{\text{phys}}} \longrightarrow \mathrm{Prim}_{\Gamma_{\text{gram}}}$$

compatible with brackets. Without this, the compact Hall/factorization target cannot literally be $\mathfrak{g}_{\Delta_5}$-graded by (n, l, m).

2. The “half K3 index” is not physical data

The draft says:

$$Z_{K3} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n,l) q^n r^l.$$

Then it introduces

$$\mathcal{U}_{n,l} \in K_0(\mathrm{sVect}_{fd}), \qquad \mathrm{sdim}\, \mathcal{U}_{n,l} = f(n,l),$$

and correctly says this is not a canonical half of the K3 BPS Hilbert space.

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That is honest, but it also means the phrase “one-particle index” is dangerous. A hostile physicist will object:

$$\mathrm{sdim}\, H^{\mathrm{BPS}}_{n,l} = 2f(n,l)$$

is an index of the K3 sigma model. The half-index $f(n,l)$ is an arithmetic input to the Borcherds lift. It is not a protected subspace, not a superselection sector, and not a Hilbert-space trace.

The situation is even sharper for negative coefficients. If $f(n,l) < 0$, your model takes an odd vector space of dimension $-f(n,l)$. That is a perfectly valid K_0-representative.

But it is arbitrary. Many parity splits have the same superdimension. The determinant only sees

$$\dim V_0 - \dim V_1,$$

not $(\dim V_0, \dim V_1)$. Your own Proposition 10.29/10.30 says the denominator product does not determine parity dimensions, zero-superdimension spaces, or structure constants. raeez.lorgat.automorphic-correc...

Required repair

Stop calling $\mathcal{U}_{n,l}$ a "half-index object" without immediately saying:

> This is an arbitrary Grothendieck representative of the arithmetic half-index. It has no physical Hilbert-space status.

Rename

$$Z^{\mathrm{vir}}_{\mathrm{BPS}}$$

to something like

$$Z^{\mathrm{vir}}_{\mathrm{Borch}}$$

unless and until a compact BPS realization is actually constructed.

3. The main determinant theorem is mostly an imported product theorem

The paper's determinant proof is:

1. Choose a K_0-graded virtual object with superdimensions $f(nm, l)$.
2. Apply the formal Fock/superdeterminant identity.
3. Invoke the Gritsenko–Nikulin Borcherds product to identify the result with Δ_5.

This is mathematically valid, but it is not a new construction of Δ_5, nor of a BPS determinant. Gritsenko–Nikulin already give the product

$$\frac{1}{64}\Delta_5(Z) = e^{\pi i(z_1+z_2+z_3)} \prod_{(n,l,m)>0} \left(1 - e^{2\pi i(nz_1+lz_2+mz_3)}\right)^{f(nm,l)}.$$

That is exactly the spine of your determinant. arXiv +1

So the contribution is not “we derive the Igusa square root.” The contribution is:

> We package the known Borcherds product as a virtual K_0-superdeterminant, fix the theta-vs-monic normalization, and separate it from the OP scalar square and the open compact realization problem.

That is a respectable paper. It is just a more modest one.

Required repair

Change theorem names like “Virtual protected determinant” and “main synthesis theorem” so that they do not sound like new physical construction. Use titles like:

> Normalized K_0-determinant form of the Gritsenko–Nikulin product.

4. The OP/DT scalar square is a scalar theorem, not evidence for the square root

The OP/DT part is externally legitimate: Oberdieck–Pixton prove the primitive $K3 \times E$ Igusa cusp form conjecture in the reduced Gromov–Witten setting, and Oberdieck’s reduced stable-pair work compares quotient Behrend-weighted definitions and reduced virtual-class definitions. arXiv +1

But the physics/mathematics relation is easy to overstate. Bryan’s account makes the scalar conjecture

$$DT(X) = -\frac{1}{\chi_{10}}$$

with the monic product convention for χ_{10}, and also emphasizes that ordinary DT invariants vanish because of the E-action, forcing a reduced quotient theory.

UBC Mathematics

Your conversion

$$\chi_{10}^{OP} = D_5^2 = (64^{-1}\Delta_5)^2, \qquad Z_{OP}^X = -4096\Delta_5^{-2}$$

is fine as a normalization statement. But it has no physical content beyond normalization. The number 4096 is not a degeneracy, not an anomaly, not an orientation character, and not a square-root datum. You say this, but the scalar-square section is still positioned so prominently that readers may infer more.

Required repair

Move the scalar square after the denominator algebra, not before the physical interpretation. Present it as:

> The OP theorem sees D_5^2, not D_5. Therefore it confirms the scalar square only in the reduced enumerative theory. It supplies no square-root orientation.

5. Your “primitive recognition theorem” is nearly tautological

The compact recognition theorem assumes:

- oriented Hall object,
- protected integration,
- chiral bar-cobar comparison,
- primitive representatives,
- parity split,
- Chevalley relations,
- real Serre relations,
- imaginary orthogonality relations,
- no-extra-relations theorem,
- completed PBW compatibility,
- same radical quotient,
- full root-space parity dimensions.

Then it concludes:

$$H^0\mathrm{Prim}^{prot}(F_X) \cong \mathfrak{g}_{\Delta_5}.$$

This is formally true, but a hostile reader will call it a restatement of the desired isomorphism. Your Definition 10.19 already packages essentially everything required for the conclusion. raeez.lorgat.automorphic-correc...

The theorem is useful as a **certificate checklist**. It is weak as a theorem.

Required repair

Rename it:

> Recognition certificate for a compact Igusa realization.

Then state:

> This is not a theorem producing new structure. It is a list of data whose verification would be sufficient to identify a compact primitive algebra with $\mathfrak{g}_{\Delta_5}$.

That will make it harder to attack.

6. The E_3-factorization language is underdefined

You repeatedly refer to a "holomorphic E_3-factorization algebra" on the Calabi–Yau threefold $K3 \times E$. You also say the symbol is a placeholder and not a constructed functor. raeez.lorgat.automorphic-correc…

That is good honesty, but the phrase itself is not standard enough to carry the argument. In a real six-dimensional topological theory one expects E_6-type local structure. In a holomorphic theory on a complex threefold one gets a holomorphic factorization algebra, but not automatically an ordinary little-disks E_3-algebra unless a precise holomorphic/formality/framing reduction is supplied. You list "formality point" and "S^3-framing witness," but this is currently a slogan, not a definition.

Hostile objection

"What exactly is an E_3-algebra on a complex threefold here? Is this holomorphic disks, real little 3-disks, Costello–Gwilliam factorization, E_3 from complex dimension three, or a framed E_3 after integrating over K3? Which operad acts, and on what category?"

Required repair

Define the exact structure before using the symbol:

$$A_X \in \mathrm{Alg}_{\mathcal{O}}(\mathrm{Fact}_{hol}(X))$$

for some named operad or factorization category. If "E_3" is only heuristic, remove it from theorem-level statements and say "holomorphic factorization algebra on X" until the operadic structure is specified.

7. The $K3 \to E$ specialization is the real missing construction

The paper wants

$$F_X \simeq \mathrm{Sp}^{ch}_{K3,E}(A_X^{E_3}) \in \mathrm{Fact}(\mathrm{Ran}(E)).$$

But $\mathrm{Sp}^{ch}_{K3,E}$ is not a known functor in the paper. It is a desired operation. You admit that construction is open. raeez.lorgat.automorphic-correc...

The attack is simple: all compact physics is hidden inside the undefined functor.

A reader can accept the Borcherds product without accepting any of the chiral/factorization narrative. The paper should therefore not let the undefined specialization appear as if it explains the product.

Required repair

Make the logical direction one-way:

$$\text{Known automorphic denominator} \quad \Rightarrow \quad \text{target constraints for any future } K3 \to E$$

Do not suggest that the specialization is part of the proof.

8. The Ran(E) obstruction is good, but phrase it narrowly

Your observation is correct: if a one-dimensional subscheme $C \subset K3 \times E$ has positive elliptic degree, then $p_E(C) = E$. Therefore such a curve cannot be localized at finitely many points of E. The manuscript uses this to argue that naive support-local factorization on $\mathrm{Ran}(E)$ sees only projection-finite sectors.

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That is one of the strongest original conceptual points in the draft.

But do not overstate it as a general obstruction to locality. It is an obstruction to a **particular pushforward-to-Ran(E)** model. A six-dimensional local theory on $K3 \times E$ can still have local observables; the problem is that positive elliptic-degree curve sectors are global from the E-projection viewpoint.

Required repair

Replace “locality obstruction” with:

projection-to-E support-locality obstruction.

That avoids a physicist objecting that you are confusing spacetime locality with support-locality after dimensional reduction.

9. The BKM denominator is not a microscopic BPS algebra

The denominator algebra is real mathematics: Gritsenko–Nikulin construct a generalized Lorentzian Kac–Moody superalgebra whose denominator is the normalized Δ_5 form. Their paper explicitly gives Δ_5 as the correcting automorphic form for the rank-three Cartan matrix with entries 2 on the diagonal and -2 off the diagonal. arXiv

But the BKM algebra is not automatically the algebra of BPS states. You already know this and state it. The determinant only fixes signed supermultiplicities:

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,0} - \dim \mathfrak{g}_{\alpha,1}.$$

It does not fix the two dimensions separately, invisible zero-superdimension root spaces, or brackets. raeez.lorgat.automorphic-correc...

A very simple adversarial counterexample is already implicit in your paper:

Take any graded super vector space B with

$$\mathrm{sdim}\, B_{n,l,m} = f(nm, l)$$

and give it the zero bracket. It has the same determinant, but it is not $\mathfrak{g}_{\Delta_5}$. Your own Proposition 10.31 says the determinant forgets brackets, parity splits, simple-root representatives, relation constants, and Hopf-pairing radicals.

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Required repair

Every time you write

$$\mathrm{smult}\, \mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm, l),$$

immediately distinguish:

$$\text{signed root Euler characteristic} \neq \text{state space} \neq \text{primitive Hall bracket}.$$

You do this in places. Do it everywhere.

10. The physical duality group is underspecified

The paper uses $Sp_4(\mathbb{Z})$ as the automorphic group of Δ_5 and then discusses physical charge invariants. That is fine automorphically, but physically $K3 \times E$ type II compactification has a much larger charge/duality structure. $Sp_4(\mathbb{Z})$ acts naturally on

genus-two chemical potentials and the triple of T-duality invariants, not on the full microscopic charge lattice.

So the paper needs to say:

$$Sp_4(\mathbb{Z})$$

is the modular group of the genus-two counting function / invariant chemical potential, not the full physical U-duality group of the compactification.

Otherwise a physicist will object that the true BPS charge lattice and wall-crossing data contain more information than $(Q^2/2, Q \cdot P, P^2/2)$. Multi-centered wall crossing, for example, depends on splittings and symplectic pairings of constituents, not only the final Gram matrix.

Required repair

Add a paragraph:

> The $Sp_4(\mathbb{Z})$ action here is the arithmetic symmetry of the genus-two automorphic form and of the reduced invariant variables. It is not asserted to be the full U-duality action on the microscopic charge lattice.

11. The $m = 0$ boundary factors are not one-particle K3 states

You define

$$V_{n,l,m} = \mathcal{U}_{nm,l}.$$

For $m > 0$, this matches the Borcherds/DMVV-style exponent $f(nm, l)$. But for $m = 0$, you get infinitely many boundary factors with the same K3 coefficient $f(0, l)$ as n varies. Those are cusp/Weyl/oscillator boundary factors in the product. They are not literal copies of a K3 BPS Hilbert space indexed by arbitrary n.

You note that $m = 0$ are cusp-boundary terms, but the “one-particle object” language still invites a wrong physical reading.

Required repair

Split the virtual object:

$$V = V_{\text{bulk}} \oplus V_{\text{cusp}}.$$

Then state:

- V_{bulk} is the Borcherds/Hecke exponent package for $m > 0$;
- V_{cusp} is a boundary/Weyl correction package;
- neither is a microscopic one-particle Hilbert space.

12. Boundary rows: good ledger, but too much row-map danger

The Cléry–Gritsenko result really is an eight-row classification of genus-two Siegel modular forms with the simplest diagonal divisor. arXiv

Govindarajan's CHL paper also states that the square roots of the relevant twisted dyon Siegel forms are precisely the dd-modular forms classified by Cléry–Gritsenko and arise as Weyl–Kac–Borcherds denominators. arXiv

So your ledger is defensible. But this part is high-risk because you are mixing:

- Cléry–Gritsenko row order,
- your row order,
- CHL $G_N(M)$ notation,
- product-degree lattices,
- multiplier covers,
- scalar squares,
- denominator identities,
- physical hosts.

A hostile reader will search for one row-map mismatch and use it to discredit the appendix.

Required repair

Move the eight-row discussion to a separate paper or a clearly optional appendix. If kept, every row needs a compact table with:

$$(F_j, \Gamma_j, \mathrm{cover}, \nu_j, \mathrm{seed}, \rho_j, L_j, \mathrm{source\ theorem}).$$

For $F_7 = \nabla_{3/2}$, do not merely say "μ_4-cover." Explicitly state which multiplier is pulled back, what becomes single-valued, and which square-root branch is used.

13. The L-function appendix is a distraction

The spin L-factor normalization for Δ_5^2 is not needed for the main paper. Even if correct, it invites a referee to audit a completely separate Saito–Kurokawa normalization chain.

Your main argument does not require:

$$L_{\mathrm{spin}}(s, \Delta_{10}) = \zeta(s-8)\zeta(s-9)L(s, \Delta E_6).$$

It does not help construct the square root, the Hall orientation, the compact factorization algebra, or the primitive bracket. It is a liability.

Required repair

Remove it, or move it to a separate note titled “normalization appendix, independent of the main theorem.”

14. The current-envelope construction is formal, not geometric

You define

$$\mathrm{Cur}_E(A_{\mathrm{den}}) = A_{\mathrm{den}} \otimes \mathcal{O}_E$$

and then

$$\mathrm{Den}_{\Delta_5,E} = U_E^{ch}(\mathrm{Cur}_E(A_{\mathrm{den}})).$$

This is fine as a formal Beilinson–Drinfeld current algebra. But it is not a compact BPS chiral algebra. The construction works for any Lie superalgebra; it does not use $K3 \times E$ geometry except by naming E.

Your paper says this, but the notation $\mathrm{Den}_{\Delta_5,E}$ may still make readers think the curve E has acquired physical content.

Required repair

Call it:

$$\mathrm{CurEnvelope}_E(\mathfrak{g}_{\Delta_5})$$

and say:

> This is a formal target object, not a result of compactification.

15. The strongest version of the paper is much simpler

I would restructure the manuscript around a cleaner thesis:

Theorem A: normalization

$$D_{\phi_{0,1}}(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

This is the theta-normalized Gritsenko–Nikulin product.

Theorem B: denominator target

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1} \Delta_5(2Z), \qquad \operatorname{smult} \mathfrak{g}_{\alpha(n,l,m)} = f(nm, l)$$

on active support.

Theorem C: scalar square normalization

$$Z^X_{OP/DT} = -4096 \Delta_5^{-2} = -(64)^2 D_{\phi_{0,1}}^{-2}.$$

Open Problem D: compact realization

Construct a real additive physical charge lattice, Hall category, orientation, protected integration, chiral specialization, and primitive bracket whose invariant pushforward recovers the BKM denominator.

This framing would be almost unassailable.

Bottom-line attack

The paper is mathematically credible where it says:

$$\text{Borcherds product of } \phi_{0,1} = \Delta_5, \qquad \text{GN denominator } = 64^{-1} \Delta_5(2Z), \qquad \text{OP scal}$$

It becomes vulnerable where it lets the reader believe:

$$\Delta_5^{-1}$$

is a constructed compact BPS partition function, or that

$$(n, l, m)$$

is already a physical Hall charge. It is not. It is a Fourier/root degree. The physical charge lattice problem is the central unresolved issue.

Your best defense is to lean into that: make the paper a precise **no-overclaim ledger**. The title could remain provocative, but the theorem statements should be austere. The compact $K3 \times E$ realization should be framed as a recognition problem constrained by the automorphic data, not as something the determinant has substantially built.

Sources

Sources

No you don't seem to get it.

The point is NOT to cut the paper down to size. it's to understand what it is trying to prove and produce those constructions, derivations, deductions & mathematics

Show more

Let's construct, deduct, prove the The physical charge lattice problem which is the central unresolved issue.

You're right. I'll switch from referee containment to constructive reconstruction: identify the intended compact Igusa realization and build the missing mathematical and physical machinery as far as possible.

The first construction should be a two-layer charge formalism: additive microscopic charges upstairs, additive Gram-current degrees downstairs, linked by a non-additive moment map rather than quotienting physics away.

I'll aim to prove recovery of the Igusa/BKM $\mathbb{Z}^3$-grading as an invariant pushforward or sector degree, with convolution induced by block-sum/orthogonal decompositions.

Thought for 11m 24s

Yes. Let's build the missing object.

The correct resolution is:

$$\Gamma_{\text{phys}} \neq \Gamma_{\text{ind}}.$$

The physical Hall/brane charge lattice is additive. The Igusa/Borcherds lattice $\Gamma_{\text{ind}} \cong \mathbb{Z}^3$ is the **rank-two Gram invariant** of that additive lattice. The construction is not to force (n, l, m) to be microscopic; it is to prove that (n, l, m) is the invariant shadow of a real additive physical charge lattice, and then define the correct descent.

Your paper already has the ingredients: the D6–D2–D0 branch, the OP variables, the Gram convention, and the statement that Γ_{ind} is not itself microscopic. The missing construction is the **dyonic Mukai lattice over $K3 \times E$.** raeez.lorgat.automorphic-correc...

1. The physical charge lattice

Let S be the K3 surface and $X = S \times E$. Put

$$\Lambda_S := H(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

with Mukai pairing

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's.$$

This is the K3 Mukai lattice. It is even, unimodular, and has signature $(4, 20)$. In the full $N = 4$ charge theory one may enlarge this to the Narain lattice $\Gamma^{6,22}$, but for the algebraic D-brane sector of $K3 \times E$, $H(S, \mathbb{Z})$ is the correct geometric starting point. Springer Link

Choose the standard basis

$$1_E \in H^0(E, \mathbb{Z}), \qquad \omega_E \in H^2(E, \mathbb{Z}), \qquad \int_E \omega_E = 1.$$

Since $H^{\text{odd}}(S) = 0$,

$$H^{\text{even}}(S \times E, \mathbb{Z}) \cong H(S, \mathbb{Z}) \otimes H^{\text{even}}(E, \mathbb{Z}).$$

Therefore every even RR/Mukai charge can be written uniquely as

$$v_X(F) = P(F) \otimes 1_E + Q(F) \otimes \omega_E,$$

with

$$P(F), Q(F) \in \Lambda_S.$$

Thus define the **dyonic Mukai charge lattice**

$$\Gamma_X^{\mathrm{phys}} := \Lambda_S \oplus \Lambda_S.$$

An object F has physical charge

$$\mathrm{ch}_{\mathrm{phys}}(F) = (Q(F), P(F)) \in \Gamma_X^{\mathrm{phys}}.$$

This is the additive lattice. For an exact triangle or Hall extension

$$F_1 \to F \to F_2,$$

one has

$$Q(F) = Q(F_1) + Q(F_2), \qquad P(F) = P(F_1) + P(F_2).$$

So the Hall algebra must be graded by

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S,$$

not directly by

$$\Gamma_{\mathrm{ind}} = \mathbb{Z}^3.$$

2. The Gram-index map

Now define the Gram map

$$\Pi : \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_{\mathrm{ind}} \cong \mathbb{Z}^3$$

by

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

Because Λ_S is even,

$$\frac{Q^2}{2}, \frac{P^2}{2} \in \mathbb{Z}, \qquad Q \cdot P \in \mathbb{Z}.$$

Equivalently, define

$$T(Q,P) = \begin{pmatrix} Q^2/2 & Q\cdot P/2 \\ Q\cdot P/2 & P^2/2 \end{pmatrix}.$$

Then

$$2T(Q,P) = \begin{pmatrix} Q^2 & Q\cdot P \\ Q\cdot P & P^2 \end{pmatrix}.$$

This is exactly the matrix your paper calls the physical Gram matrix for a protected electric-magnetic pair. raeez.lorgat.automorphic-correc...

The discriminant is

$$\Delta_{\mathrm{gram}}(Q,P) = 4\begin{matrix} Q^2 & P^2 \\ 2 & 2 \end{matrix} - (Q\cdot P)^2 = Q^2P^2 - (Q\cdot P)^2.$$

So the Borcherds variables are not arbitrary formal variables. They are the Fourier characters of the rank-two Gram form:

$$q^n r^l s^m = \exp\bigl(2\pi i \,\mathrm{tr}(T(Q,P)Z)\bigr),$$

where

$$(n,l,m) = \Pi(Q,P).$$

3. The D6–D2–D0 computation

Now prove that this construction reproduces your OP/DT variables exactly.

Let $Y \subset X = S \times E$ be a one-dimensional subscheme with

$$[Y] = (\beta, d) \in H_2(S,\mathbb{Z}) \oplus H_2(E,\mathbb{Z}),$$

and

$$\chi(\mathcal{O}_Y) = k.$$

Here d is the elliptic degree and β is the K3 curve class. In your paper's primitive branch,

$$\beta = \beta_h = s_1 + hF, \qquad \beta_h^2 = 2h-2.$$

The ideal sheaf I_Y has

$$\mathrm{ch}(I_Y) = 1 - \mathrm{PD}[Y] - k[\mathrm{pt}_X].$$

Because

$$\mathrm{td}(S) = (1, 0, 1), \qquad \mathrm{td}(E) = 1,$$

the Mukai vector is

$$v_X(I_Y) = \mathrm{ch}(I_Y) \quad \mathrm{td}(X).$$

Now decompose $\mathrm{PD}[Y]$. The K3-vertical curve $\beta \times \{\mathrm{pt}\}$ contributes

$$\beta^\vee \otimes \omega_E,$$

and the elliptic curve class $d\,\{\mathrm{pt}\} \times E$ contributes

$$d[\mathrm{pt}_S] \otimes 1_E.$$

Therefore

$$v_X(I_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta, -k) \otimes \omega_E.$$

So set

$$P_Y = (1, 0, 1 - d), \qquad Q_Y = (0, -\beta, -k).$$

Then

$$Q_Y^2 = (-\beta)^2 = \beta^2 = 2h - 2,$$

so

$$\frac{Q_Y^2}{2} = h - 1.$$

Also

$$P_Y^2 = 0 - 2(1)(1 - d) = 2(d - 1),$$

so

$$\frac{P_Y^2}{2} = d - 1.$$

Finally,

$$Q_Y \cdot P_Y = (-\beta) \cdot 0 - 0(1 - d) - 1(-k) = k.$$

Hence

$$\Pi(Q_Y, P_Y) = (h-1,\ k,\ d-1).$$

This is exactly the OP/Borcherds variable dictionary:

$$q_{\mathrm{Bch}}^{h-1} r_{\mathrm{Bch}}^{k} s_{\mathrm{Bch}}^{d-1}.$$

So the shifts $h-1$ and $d-1$ are not ad hoc. They are the Mukai self-pairings of the two Künneth components of the D6–D2–D0 charge.

In particular,

$$\text{K3 norm} = \frac{Q_Y^2}{2} = h-1,$$

$$\text{D0/Fourier index} = Q_Y \cdot P_Y = k,$$

$$\text{elliptic degree} = \frac{P_Y^2}{2} = d-1.$$

This gives the physical derivation of your (n, l, m) variables.

4. The additive defect and its exact cocycle

Now prove why Γ_{ind} is not the Hall charge lattice.

Take two physical charges

$$c_1 = (Q_1, P_1), \qquad c_2 = (Q_2, P_2).$$

The physical sum is

$$c_1 + c_2 = (Q_1 + Q_2,\ P_1 + P_2).$$

Then

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where the cross-term is

$$B(c_1, c_2) = (Q_1 \cdot Q_2,\ Q_1 \cdot P_2 + Q_2 \cdot P_1,\ P_1 \cdot P_2)\,.$$

Proof:

$$\frac{(Q_1+Q_2)^2}{2} = \frac{Q_1^2}{2} + \frac{Q_2^2}{2} + Q_1 \cdot Q_2,$$

$$(Q_1+Q_2)\cdot(P_1+P_2) = Q_1\cdot P_1 + Q_2\cdot P_2 + Q_1\cdot P_2 + Q_2\cdot P_1,$$

$$\frac{(P_1+P_2)^2}{2} = \frac{P_1^2}{2} + \frac{P_2^2}{2} + P_1\cdot P_2.$$

Thus

$$\Pi \text{ is quadratic, not additive.}$$

This is not a failure. It is the key structural fact.

The Hall algebra is graded by

$$\Gamma_X^{\mathrm{phys}}.$$

The Igusa/Borcherds product is graded by the invariant shadow

$$\Gamma_{\mathrm{ind}}.$$

The descent from the former to the latter must account for the cocycle B.

5. The Gram-corrected charge extension

We can package the defect as an honest additive object.

Define

$$\Gamma_X := \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{ind}}.$$

For $c, c' \in \Gamma_X^{\mathrm{phys}}$ and $T, T' \in \Gamma_{\mathrm{ind}}$, define

$$(c,T)\star(c',T') = \big(c+c',\ T+T'+B(c,c')\big).$$

Because B is symmetric bilinear, this product is commutative and associative. Indeed,

$$B(c+c',c'') = B(c,c'') + B(c',c''),$$

so

$$((c,T)\star(c',T'))\star(c'',T'') = (c+c'+c'',\ T+T'+T''+B(c,c')+B(c,c'')+B(c',$$

and the same expression is obtained from

$$(c, T) \star ((c', T') \star (c'', T'')).$$

The identity is $(0, 0)$. The inverse of (c, T) is

$$(-c,\ -T + B(c, c)).$$

Now define

$$s : \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_X$$

by

$$s(c) = (c, \Pi(c)).$$

Then

$$s(c) \star s(c') = (c, \Pi(c)) \star (c', \Pi(c')) = (c + c', \Pi(c) + \Pi(c') + B(c, c')) = (c + c', \Pi(c$$

Therefore

$$s : \Gamma_X^{\mathrm{phys}} \hookrightarrow \Gamma_X \text{ is an additive embedding.}$$

This is the precise mathematical repair of the non-additivity problem.

The physical charge lattice remains Γ_X^{phys}. The Gram-index lattice becomes a quadratic invariant, and the obstruction to additivity is the explicit bilinear cocycle B. The central extension Γ_X is the minimal bookkeeping object that makes the Gram data functorial under physical charge addition.

6. Primitive torsion-one descent

Now we need the invariant quotient.

Let

$$\Lambda = \Lambda_S$$

or, in the full $N = 4$ theory,

$$\Lambda = \Gamma^{6,22}.$$

Let

$$c = (Q, P) \in \Lambda \oplus \Lambda.$$

Define the rank-two charge sublattice

$$M_c := \mathbb{Z}Q + \mathbb{Z}P \subset \Lambda.$$

The Gram form on M_c is

$$G_c = \begin{pmatrix} Q^2 & Q \cdot P \\ Q \cdot P & P^2 \end{pmatrix} = 2T(\Pi(c)).$$

The pair $c = (Q, P)$ is **primitive saturated**, or torsion-one, if

$$M_c = \Lambda \cap (M_c \otimes \mathbb{Q}).$$

Equivalently, in a coordinate basis this is the condition usually written

$$\gcd(Q \wedge P) = 1.$$

The original DVV Igusa-counting formula is the torsion-one dyon formula; torsion > 1 dyons require extra discrete data beyond $Q^2, P^2, Q \cdot P$. arXiv +1

Now use the lattice theorem.

For an even unimodular indefinite lattice Λ with enough hyperbolic planes, primitive embeddings of a fixed even rank-two lattice M are unique up to $O(\Lambda)$ once the complement has sufficiently large rank relative to the discriminant length. The standard Nikulin criterion gives uniqueness when the positive and negative signatures have room and

$$\mathrm{rk}(\Lambda) - \mathrm{rk}(M) \geq 2 + \ell(A_M),$$

where $A_M = M^\vee / M$ is the discriminant group. For rank-two M, $\ell(A_M) \leq 2$, while $H(S, \mathbb{Z})$ has rank 24 and $\Gamma^{6,22}$ has rank 28, so the condition is overwhelmingly satisfied away from the usual boundary degeneracies. math.brown.edu

Thus, for primitive saturated pairs,

$$O(\Lambda) \backslash \{(Q, P) \in \Lambda^2 : \Pi(Q, P) = (n, l, m),\ \gcd(Q \wedge P) = 1\}$$

is a single orbit, subject to the standard signature and chamber conditions.

Therefore the protected index in the primitive torsion-one sector descends:

$$\Omega(Q,P) = \Omega(n,l,m), \qquad (n,l,m) = \Pi(Q,P).$$

This is the mathematical meaning of the Igusa variables:

$$\Gamma_{\mathrm{ind}} = \mathrm{Sym}^2(\mathbb{Z}^2)_{\mathrm{half}} \text{ is the primitive T-duality-invariant quotient of } \Gamma_X^{\mathrm{phys}}.$$

For torsion > 1, this is false unless one augments Γ_{ind} by the torsion invariant.

7. The Hall object should be rewritten

The compact Hall object should not be assumed to have numerical lattice Γ_{ind}. It should be:

$$H_X = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} H_{X,c}, \qquad \Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S.$$

Then define the Gram pushforward only after taking the protected, duality-invariant, primitive/torsion-one sector:

$$(\Pi_* H_X)_T := \bigoplus_{\substack{[c] \in O(\Lambda_S)\backslash \Gamma_X^{\mathrm{phys}} \\ \Pi(c) = T}} H_{X,c}, \qquad T = (n,l,m).$$

In the primitive saturated sector, the orbit theorem reduces this to one orbit:

$$(\Pi_* H_X)_{(n,l,m)} \simeq H_{X,c} \quad \text{for any } c \text{ with } \Pi(c) = (n,l,m).$$

Then the desired primitive protected statement becomes

$$\mathrm{sdim}\, H^0\, \mathrm{Prim}^{\mathrm{prot}} (\Pi_* F_X)_{(n,l,m)} = f(nm, l).$$

This is the correct replacement for saying directly that the compact Hall category is Γ_{ind} -graded.

The actual Hall bracket lives first as

$$[\,,\,] : \mathrm{Prim}^{\mathrm{prot}}_{c_1} \otimes \mathrm{Prim}^{\mathrm{prot}}_{c_2} \longrightarrow \mathrm{Prim}^{\mathrm{prot}}_{c_1+c_2},$$

with

$$c_i \in \Gamma_X^{\mathrm{phys}}.$$

Only after Gram descent does one compare with the BKM bracket:

$$[\,,\,]_{\mathrm{BKM}} : \mathfrak{g}_{\alpha(n,l,m)} \otimes \mathfrak{g}_{\alpha(n',l',m')} \longrightarrow \mathfrak{g}_{\alpha(n+n',l+l',m+m')}.$$

The price of descent is exactly the cocycle $B(c_1, c_2)$. So the compact realization must prove one of the following:

$$B(c_1, c_2) = 0$$

for the primitive collision channels contributing to the BKM bracket, or else it must prove that the Hall bracket descends through the Gram-corrected extension Γ_X and then collapses to the BKM root grading.

This is now a precise mathematical problem, not a vague defect.

8. What this proves for the whitepaper

The physical charge lattice problem can now be stated as a theorem.

Theorem: Physical charge lattice and Gram descent for $K3 \times E$

Let $X = S \times E$, with S a K3 surface. The additive physical D-brane charge lattice in the even RR sector is

$$\Gamma_X^{\mathrm{phys}} = H(S, \mathbb{Z}) \oplus H(S, \mathbb{Z}).$$

The Igusa index lattice

$$\Gamma_{\mathrm{ind}} \cong \mathbb{Z}^3$$

is the image of the quadratic Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

For a rank-one D6–D2–D0 ideal-sheaf charge with curve class (β, d), Euler characteristic k, and $\beta^2 = 2h - 2$,

$$v_X(I_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta, -k) \otimes \omega_E.$$

Choosing the Igusa electric-magnetic pair

$$Q_Y = (0, -\beta, -k), \qquad P_Y = (1, 0, 1-d),$$

one obtains

$$\Pi(Q_Y, P_Y) = (h-1, k, d-1).$$

Thus the OP/Borcherds monomial is exactly

$$q^{h-1} r^k s^{d-1} = \exp\big(2\pi i \operatorname{tr}(T(Q_Y, P_Y) Z)\big).$$

The map Π is not additive. Its failure is the explicit bilinear cocycle

$$B((Q_1, P_1), (Q_2, P_2)) = (Q_1 \cdot Q_2,\ Q_1 \cdot P_2 + Q_2 \cdot P_1,\ P_1 \cdot P_2).$$

The Gram-corrected extension

$$\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{ind}}$$

with multiplication

$$(c, T) \star (c', T') = (c + c', T + T' + B(c, c'))$$

makes the section

$$c \mapsto (c, \Pi(c))$$

additive. Hence the Hall category is physically graded by Γ_X^{phys}, while the Igusa/Borcherds denominator is the primitive torsion-one $O(\Lambda_S)$-invariant Gram descent of that grading.

9. The exact paper-level replacement

Replace the sentence

$$\text{“Suppose an extension-closed compact Hall category with numerical Gram-index la}$$

by:

$$\text{“Suppose an extension-closed compact Hall category graded by } \Gamma_X^{\mathrm{phys}} = H(S, \mathbb{Z}) \oplus$$

Then the recognition target becomes

$$H^0 \Pi_* \operatorname{Prim}^{\mathrm{prot}}(F_X) \cong \mathfrak{g}_{\Delta_5}.$$

Not

$$H^0\,\mathrm{Prim}^{\mathrm{prot}}(F_X) \cong \mathfrak{g}_{\Delta_5}$$

before descent.

That is the crucial correction.

10. The physics interpretation

The two Künneth Mukai components

$$Q = (0, -\beta, -k), \qquad P = (1, 0, 1-d)$$

are the two physical charge vectors whose Gram invariants are counted by the Igusa variables.

The K3 curve norm gives

$$Q^2/2 = h - 1.$$

The D0/Fourier charge gives

$$Q \cdot P = k.$$

The elliptic degree, with the Todd/vacuum shift, gives

$$P^2/2 = d - 1.$$

So the Igusa triple is physically

$$(n, l, m) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2}\right) = (h-1, k, d-1).$$

This is exactly the same triple that appears in the Borcherds product

$$\prod_{(n,l,m)>0} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The product does not live directly on the raw physical lattice. It lives on the primitive Gram quotient of the physical lattice.

That is the construction.

Sources

Sources

I want it to be a dirac-inspired paper (as if it was written by dirac) on the physics here, that makes natural the factorisation algebra from physical first-principles etc. locality (or lack thereof) is treated physically. unitarity or lack thereof...

I'll recast it in a Dirac style: begin with observables, charges, locality, and the protected supercharge, then derive factorization, Hall structure, and the Igusa square root.

The crucial shift is that charged BPS states are not local fields; locality governs protected observables and collision laws before charge completion and trace reveal the automorphic product.

Thought for 6m 51s

Yes. The paper should be **Dirac-inspired** in exactly this sense:

> Do not begin with a scalar partition function and ask for a square root.
> Begin with a first-order physical object whose scalar shadow is a square.

That is the Dirac move. Dirac did not take an analytic square root of the Klein–Gordon operator. He introduced new algebraic degrees of freedom — spinors and Clifford multiplication — such that the square became scalar. Your paper should do the same. The Igusa square root should not be presented as a branch of Δ_5^2. It should be presented as the **first-order protected chiral/Pfaffian object** whose scalar reduced DT/OP theory is the square. Your current manuscript already gestures at this through protected locality, the E_3-source, the orientation character, and the statement that the scalar square forgets the automorphic reflection sign. The rewrite should make that the central physics rather than a caveat. raeez.lorgat.automorphic-correc...

Here is the spine I would build.

The Dirac principle for the Igusa square root

The scalar theory gives

$$Z^X_{\text{scalar}} = -4096\,\Delta_5(Z)^{-2}.$$

This is the analogue of the second-order equation. It is invariant under the sign character because the sign has been squared away.

The first-order theory is not

$$Z^X_{\text{scalar}}$$

as an analytic function. It is a protected chiral object

$$\mathcal{A}_X$$

with a determinant/Pfaffian

$$\mathrm{Pf}_{\text{prot}}(\mathcal{A}_X) = \Delta_5(Z),$$

and whose scalar norm or doubled trace gives

$$Z^X_{\text{scalar}} = C_{\text{scalar}}\ \mathrm{Pf}_{\text{prot}}(\mathcal{A}_X)^{-2}.$$

For the OP/DT normalization,

$$C_{\text{scalar}} = -4096.$$

Thus the actual problem is not:

$$\text{choose a square root of } \Delta_5^2.$$

It is:

$$\text{construct the first-order protected algebra whose determinant is } \Delta_5.$$

This is the Dirac formulation.

The new degrees of freedom replacing “spinors” are:

$$\text{orientation line} + \text{primitive Hall bracket} + \text{chiral/factorization locality} + \text{Borcherc}$$

The scalar square forgets them. The square root remembers them.

1. States are not first. Observables are first.

A charged BPS state is not a local field. It is a sector. Its charge is measured at infinity. Therefore the microscopic Hilbert space is not the natural first object.

The first object is the algebra of protected operations.

A state is a representation of that algebra. A partition function is a trace of that representation. A trace cannot reconstruct the algebra. So the paper should not begin with the scalar trace and then apologize that it cannot reconstruct locality. It should begin with locality and then show what trace it produces.

The physical hierarchy should be:

$$\text{local protected operations} \longrightarrow \text{factorization/Hall algebra} \longrightarrow \text{primitive Lie algebra}$$

This is already latent in your manuscript, but it should become the philosophical and technical backbone.

2. The physical charge lattice

Let

$$X = S \times E,$$

where S is a K3 surface.

The even Mukai lattice of S is

$$\Lambda_S = H(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}),$$

with pairing

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's.$$

Since

$$H^{\text{even}}(X, \mathbb{Z}) \cong H(S, \mathbb{Z}) \otimes H^{\text{even}}(E, \mathbb{Z}),$$

every even charge decomposes as

$$v_X(F) = P(F) \otimes 1_E + Q(F) \otimes \omega_E.$$

Thus the additive physical charge lattice is

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S,$$

with charge

$$c(F) = (Q(F), P(F)).$$

This is the lattice on which Hall multiplication is genuinely additive.

The Igusa variables are not this lattice. They are the Gram shadow:

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right) = (n,l,m).$$

So the Fourier monomial is

$$q^n r^l s^m = \exp\left(2\pi i \,\mathrm{tr}(T(Q,P)Z)\right),$$

where

$$T(Q,P) = \begin{pmatrix} Q^2/2 & Q\cdot P/2 \\ Q\cdot P/2 & P^2/2 \end{pmatrix}.$$

Thus the genus-two variable Z is a chemical potential for the rank-two charge plane spanned by Q and P.

For the D6–D2–D0 ideal-sheaf sector,

$$v_X(I_Y) = (1,0,1-d)\otimes 1_E + (0,-\beta,-k)\otimes \omega_E.$$

So

$$P_Y = (1,0,1-d), \qquad Q_Y = (0,-\beta,-k).$$

If

$$\beta^2 = 2h-2,$$

then

$$\frac{Q_Y^2}{2} = h-1,$$

$$Q_Y\cdot P_Y = k,$$

$$\frac{P_Y^2}{2} = d - 1.$$

Therefore

$$\Pi(Q_Y, P_Y) = (h - 1, k, d - 1).$$

This is the physical derivation of the Borcherds/OP variables.

The paper should state this early. It makes (n, l, m) natural without pretending that $\mathbb{Z}^3$ is the microscopic charge lattice.

3. The local object on X

The local object is not the scalar DT series. The local object is the protected open-string observable algebra on the Calabi–Yau threefold X.

In the B-model language, the local field theory on a Calabi–Yau threefold is holomorphic Chern–Simons/BV theory. Locally, for a brane background F, the fields are controlled by the dg Lie algebra

$$\mathfrak{g}_F(U) = \mathrm{RHom}_U(F, F)[1]$$

or, perturbatively,

$$\Omega_c^{0,*}(U, \mathrm{End}\, F)[1].$$

The BV pairing comes from Serre duality and the holomorphic volume form on X. The classical interaction is cubic:

$$S(A) = \int_X \Omega_X \wedge \mathrm{Tr}\left(\frac{1}{2} A\bar{\partial} A + \frac{1}{3} A^3\right).$$

The compact nonperturbative version replaces one fixed brane by the derived stack of compactly supported branes.

For an open $U \subset X$, let

$$\mathfrak{M}_c(U)$$

be the derived stack of compactly supported coherent B-branes in U with physical charge

$$c \in \Gamma_X^{\mathrm{phys}}.$$

The Calabi–Yau threefold structure gives this stack a (-1)-shifted symplectic structure. After choosing an orientation line $\mathfrak{o}_c$, define the protected local object

$$\mathcal{A}_X(U) = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} H_*^{BM}\left(\mathfrak{M}_c(U), \Phi_c \otimes \mathfrak{o}_c\right).$$

Here Φ_c denotes the vanishing-cycle/perverse-sheaf coefficient system coming from the local critical model, and the completion is the physical charge/HN completion.

This is the object the paper should call the compact protected observable algebra.

It is not yet a Hilbert space. It is a cohomological algebra of protected operations.

4. Why this is naturally an E_3-factorization algebra

The reason is purely local.

Let $U_1, \dots, U_k \subset V \subset X$ be pairwise disjoint holomorphic opens. If

$$F_i \in \mathfrak{M}_{c_i}(U_i),$$

then

$$F_1 \oplus \cdots \oplus F_k \in \mathfrak{M}_{c_1+\cdots+c_k}(V).$$

Because the supports are disjoint,

$$\mathrm{Ext}^\bullet(F_i, F_j) = 0 \qquad (i \neq j).$$

Therefore the deformation-obstruction theory splits:

$$\mathrm{RHom}\left(\bigoplus_i F_i, \bigoplus_i F_i\right) \simeq \bigoplus_i \mathrm{RHom}(F_i, F_i).$$

The critical functions add, the vanishing cycles factor by Thom–Sebastiani, and the orientation lines multiply:

$$\Phi_{\oplus_i F_i} \simeq \boxtimes_i \Phi_{F_i},$$

$$\mathfrak{o}_{\oplus_i F_i} \simeq \boxtimes_i \mathfrak{o}_{F_i}.$$

Hence one obtains factorization maps

$$\mathcal{A}_X(U_1) \otimes \cdots \otimes \mathcal{A}_X(U_k) \longrightarrow \mathcal{A}_X(V).$$

When the opens collide, the cross-Ext groups no longer vanish. Their nonzero extensions are exactly the Hall correspondences:

$$\mathfrak{M}_{c_1} \times \mathfrak{M}_{c_2} \longleftarrow \mathfrak{E}_{c_1,c_2} \longrightarrow \mathfrak{M}_{c_1+c_2}.$$

Thus disjointness gives ordinary factorization; collision gives the Hall product.

This is the physical origin of the factorization algebra.

The reason the structure is called holomorphic E_3, rather than real E_6, is that the protected supercharge has killed the anti-holomorphic translations. The remaining local singularities depend on three complex holomorphic directions. A small holomorphic polydisc in X behaves like the local model for a holomorphic E_3-algebra. With a choice of formality/framing, one obtains an ordinary E_3-shadow.

So the paper should say:

$$\mathcal{A}_X \in \mathrm{Fact}^{\mathrm{hol}}_{E_3}(X)$$

means:

> the protected local observables of the holomorphic Calabi–Yau threefold theory form a holomorphic factorization algebra whose pointwise local model is an E_3-algebra after the chosen formality/framing.

This makes the E_3 natural, not decorative.

5. $K3 \to E$ specialization

Now project

$$p_E : X = S \times E \to E.$$

For an open $V \subset E$, define the projection-finite sector

$$\mathfrak{M}^{\mathrm{pf}}_c(V) = \{F \in \mathfrak{M}_c(S \times V) \ : \ p_E(\mathrm{Supp}\, F) \subset V \text{ is finite}\}\,.$$

Then

$$\mathcal{F}_X^{\mathrm{pf}}(V) = \bigoplus_c H_*^{BM}\left(\mathfrak{M}_c^{\mathrm{pf}}(V), \Phi_c \otimes \mathfrak{o}_c\right)$$

is a genuine chiral/factorization object on $\mathrm{Ran}(E)$.

For disjoint $V_i \subset V$, projection-finite supports remain disjoint, so one has

$$\mathcal{F}_X^{\mathrm{pf}}(V_1) \otimes \cdots \otimes \mathcal{F}_X^{\mathrm{pf}}(V_k) \longrightarrow \mathcal{F}_X^{\mathrm{pf}}(V).$$

Collisions in E remember extensions inside $S \times E$. Therefore the chiral algebra on E is not imposed by hand. It is the $K3$-integrated shadow of the local E_3-object on X.

Symbolically,

$$\mathcal{F}_X^{\mathrm{pf}} \simeq \mathrm{Sp}_{S,E}^{\mathrm{ch}}(\mathcal{A}_X)_{\mathrm{pf}}.$$

This is the correct physical meaning of the $K3 \to E$ specialization.

6. Locality fails exactly where it should fail

Now the important point: positive elliptic degree is not local on E.

Let $C \subset S \times E$ be a one-dimensional subscheme with positive elliptic degree:

$$(p_E)_*[C] = d[E], \qquad d > 0.$$

Then $p_E(C)$ is a one-dimensional closed subset of the irreducible curve E. Hence

$$p_E(C) = E.$$

Therefore C cannot be assigned to finitely many points of E.

This is not a mathematical inconvenience. It is the physics.

A brane wrapping E is not a local insertion on E. It is a global sector, just as a Wilson loop is not a pointlike local observable in the direction it wraps. It can couple to local insertions, but it is not itself localized at their points.

Therefore the correct object is not simply

$$\mathrm{Fact}(\mathrm{Ran}(E)).$$

It is a hybrid object:

$$\mathcal{F}_X^{\mathrm{hyb}} = \mathcal{F}_X^{\mathrm{pf}} \curvearrowright \mathcal{W}_X.$$

Here

$$\mathcal{W}_X = \bigoplus_{m>0} \mathcal{W}_{X,m}$$

is the wrapped elliptic-degree sector, with

$$m = \frac{P^2}{2} = d - 1.$$

The projection-finite factorization algebra acts on the wrapped sector by mixed Hall correspondences:

$$\mathcal{F}_X^{\mathrm{pf}}(V_1) \otimes \cdots \otimes \mathcal{F}_X^{\mathrm{pf}}(V_k) \otimes \mathcal{W}_{X,m} \longrightarrow \mathcal{W}_{X,m'}.$$

Wrapped-wrapped extensions give

$$\mathcal{W}_{X,m} \otimes \mathcal{W}_{X,m'} \longrightarrow \mathcal{W}_{X,m+m'+\mathrm{interaction}}.$$

The interaction term is measured on the additive physical charge lattice before Gram descent.

This is the physical replacement for the naive Ran(E) locality statement.

The s-direction in the Igusa product is therefore not a local point-support direction on E. It is the wrapped/global direction. The factorization algebra survives, but it becomes **stratified**:

$$\text{local vertical sector} + \text{global wrapped sector} + \text{mixed Hall action}.$$

That is the correct locality physics.

7. The Hall bracket and primitive algebra

The extension correspondence is

$$\mathfrak{M}_c \times \mathfrak{M}_{c'} \longleftarrow \mathfrak{E}_{c,c'} \longrightarrow \mathfrak{M}_{c+c'}.$$

After orientation and protected integration, it gives a Hall product

$$* : H_c \otimes H_{c'} \longrightarrow H_{c+c'}.$$

The primitive part of the completed Hall coalgebra gives a Lie superalgebra:

$$[\,,\,] : \mathrm{Prim}_c^{\mathrm{prot}} \otimes \mathrm{Prim}_{c'}^{\mathrm{prot}} \longrightarrow \mathrm{Prim}_{c+c'}^{\mathrm{prot}}.$$

This bracket is genuinely graded by

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S.$$

The Igusa/Borcherds algebra is obtained only after Gram descent:

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right).$$

Thus the comparison should be

$$H^0 \Pi_* \mathrm{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \cong \mathfrak{g}_{\Delta_5}.$$

Not

$$H^0 \mathrm{Prim}^{\mathrm{prot}}(\mathcal{F}_X) \cong \mathfrak{g}_{\Delta_5}$$

before descent.

The descent is physical: the full charge lattice is microscopic, but the protected automorphic trace sees only the duality-invariant Gram plane.

8. The Dirac square root and the orientation character

The scalar square is insensitive to the sign character:

$$\Delta_5^2$$

is scalar.

But the square root

$$\Delta_5$$

transforms with the Maass character

$$\nu_{\Delta_5}.$$

Physically this character must come from the square root of a determinant line. That is the exact analogue of a Pfaffian.

Let $\mathscr{L}_c$ be the determinant line of the deformation complex at charge c:

$$\mathscr{L}_c = \det \mathrm{RHom}(F, F).$$

An orientation is a square root

$$\mathfrak{o}_c^{\otimes 2} \simeq \mathscr{L}_c.$$

A wall reflection preserves $\mathscr{L}_c$, but it may reverse the square root $\mathfrak{o}_c$. Therefore the scalar determinant is invariant, while the Pfaffian changes sign.

For the three type-II simple reflections

$$s_{\delta_1}, \quad s_{\delta_2}, \quad s_{\delta_3},$$

the expected local wall model has one real normal mode. Hence

$$\epsilon_o(s_{\delta_i}) = -1, \qquad i = 1, 2, 3.$$

The scalar square sees

$$(-1)^2 = 1.$$

The square root sees

$$-1.$$

Therefore the orientation-monodromy character should satisfy

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

This is not a normalization issue. It is the physical sign of the first-order theory.

In Dirac language:

$$\Delta_5^{-2}$$

is the scalar second-order object.

$$\Delta_5^{-1}$$

is the chiral first-order object.

The passage from the second to the first requires an orientation/Pfaffian datum, just as the passage from Klein–Gordon to Dirac requires spinor/Clifford data.

9. Unitarity is used before the index, not after it

The paper should be very explicit here.

The untwisted theory is unitary. This matters because it gives an adjoint supercharge:

$$Q^\dagger,$$

and an identity of the form

$$\{Q, Q^\dagger\} = H_{\text{right}}.$$

Then all states with positive right-moving energy pair and cancel in the protected supertrace. This is why the K3 elliptic genus localizes.

So unitarity supplies the cancellation theorem.

But after taking Q-cohomology, one no longer has a positive Hilbert space. One has a cohomological, charge-graded, virtual object.

Thus

$$f(n, l)$$

is a superdimension, not a dimension.

A negative coefficient means an odd excess:

$$f(n, l) < 0 \quad \Longleftrightarrow \quad \dim H_{\bar{1}} - \dim H_{\bar{0}} > 0$$

in whatever protected representative realizes it.

The Borcherds denominator algebra is also not a unitary Hilbert-space representation. Its Cartan lattice is Lorentzian. Its multiplicities are signed supermultiplicities:

$$\operatorname{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

This is exactly what one expects from a protected cohomological theory.

So the correct statement is:

unitarity is an input to localization; the square root is a cohomological protected

That is not a weakness. It is the nature of an index.

10. The determinant as the protected first-order amplitude

Assume the primitive protected algebra exists and Gram descends to

$$\Pi_* \mathrm{Prim}^{\mathrm{prot}} = \bigoplus_{(n,l,m)} P_{n,l,m}.$$

The Dirac/IQFT requirement is

$$\mathrm{sdim}\, P_{n,l,m} = f(nm, l).$$

Then the first-order protected determinant is

$$D_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)>0} (1 - q^n r^l s^m)^{\mathrm{sdim}\, P_{n,l,m}}.$$

Hence

$$D_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)>0} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

This is the protected Pfaffian/determinant of the first-order object.

The scalar OP/DT theory gives

$$Z^X_{\mathrm{OP/DT}} = -4096\, \Delta_5^{-2} = -4096\, D_X^{-2}.$$

So the OP/DT scalar theory is the squared, orientation-forgetting theory.

This is the clean physical meaning of the Igusa square root.

11. The denominator algebra as the first-order equation

In the Dirac equation, the first-order operator is not arbitrary. Its square gives the scalar operator because the gamma matrices obey a Clifford relation.

Here the analogue is the Borcherds–Kac–Moody denominator algebra.

The primitive algebra should be generated by real and imaginary simple protected operations, with the real simple roots

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

and Cartan matrix

$$A = \begin{matrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{matrix}.$$

The first-order algebraic equation is the denominator identity

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

The scalar square follows only after taking the determinant square.

So the physical recognition theorem should be written as:

$$\text{The compact protected primitive algebra is the Dirac operator.}$$

More precisely,

$$H^0\Pi_*\mathrm{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \cong \mathfrak{g}_{\Delta_5}.$$

The denominator identity is then the first-order spectral equation of the protected algebra.

12. Proposed replacement thesis paragraph

I would put something like this near the beginning:

> The square root in this paper is meant in Dirac's sense. A scalar square is not square-rooted by choosing a branch. It is square-rooted by finding a first-order algebra whose determinant, after forgetting orientation, gives the scalar object. For $X = K3 \times E$, the scalar reduced DT/OP branch gives $-4096\Delta_5^{-2}$. The first-order object sought here is a protected holomorphic E_3-factorization algebra on X, with a hybrid chiral shadow on E, whose primitive Hall bracket Gram-descends to the Gritsenko–Nikulin Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$. The scalar square forgets the Pfaffian orientation character; the square root remembers it as the Maass character

of Δ_5. Thus the problem is not analytic extraction of Δ_5 from Δ_5^2, but construction of the protected first-order algebra whose determinant is Δ_5.

That gives the whole paper a physical mission.

13. Proposed definition: Dirac–Igusa datum

Replace the weaker "compact Igusa realization datum" with a sharper physical package.

A **Dirac–Igusa datum** for $X = S \times E$ consists of:

$$\mathfrak{D}_X = \left(\Gamma_X^{\mathrm{phys}}, \mathcal{A}_X, \mathcal{F}_X^{\mathrm{hyb}}, \mathfrak{o}, \Pi, \epsilon_o, \Theta_{\mathrm{den}}\right),$$

where:

$$\Gamma_X^{\mathrm{phys}} = H(S,\mathbb{Z}) \oplus H(S,\mathbb{Z})$$

is the additive physical charge lattice;

$$\mathcal{A}_X \in \mathrm{Fact}_{E_3}^{\mathrm{hol}}(X)$$

is the charge-completed protected local observable algebra;

$$\mathcal{F}_X^{\mathrm{hyb}} = \mathcal{F}_X^{\mathrm{pf}} \curvearrowright \mathcal{W}_X$$

is the hybrid $K3 \to E$ chiral shadow with projection-finite local sector and wrapped elliptic-degree sector;

$$\mathfrak{o}_c^{\otimes 2} \simeq \det \mathrm{RHom}(F,F)$$

is the orientation/Pfaffian datum;

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right)$$

is the Gram descent;

$$\epsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \to \{\pm 1\}$$

is the induced orientation monodromy, required to satisfy

$$\epsilon_o = \nu_{\Delta_5}$$

on the type-II Weyl group;

and

$$\Theta_{\mathrm{den}} : H^0 \Pi_* \mathrm{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \quad \sim \quad \mathfrak{g}_{\Delta_5}$$

is the primitive denominator comparison.

Then the protected determinant is

$$D_X(Z) = \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5(Z),$$

and the scalar reduced branch is

$$Z^X_{\mathrm{OP/DT}} = -4096\, D_X(Z)^{-2}.$$

This is the Dirac theorem the paper wants.

14. Locality summary in one sentence

The paper should say:

$$\text{BPS states are not local, but protected operations on BPS sectors factorize.}$$

Then refine:

$$\text{Projection-finite sectors are local on } \mathrm{Ran}(E); \quad \text{positive elliptic-degree sectors are}$$

This makes the locality obstruction physical rather than embarrassing.

15. Unitarity summary in one sentence

The paper should say:

$$\text{unitarity proves the index localization; the index itself is a cohomological amplitud}$$

Then refine:

$$\Delta_5^{-1} \text{ is orientation-sensitive and first-order;} \qquad \Delta_5^{-2} \text{ is scalar and orientation-forge}$$

That is the physics of the square root.

16. The new title-level interpretation

The title “The Igusa Square Root” then means:

$$\text{The Igusa cusp form } \Delta_5 \text{ is the Dirac/Pfaffian half of the scalar } K3 \times E \text{ index.}$$

Not half in the sense of half a Hilbert space.

Not half in the sense of an analytic branch.

Half in the sense of a first-order protected algebra whose determinant squares to the scalar branch.

That is the paper.

Sources

Sources